

Release Plan: Hylograph, Minard & Polyglot PureScript

Executive Summary

Three related projects need public release before the ShapedSteer intensive begins:

Domain	Purpose	Hosting
hylograph.net	Visualization library suite	Cloudflare Pages + Tunnel
Minard / Code Explorer	Code cartography tool (flagship app)	live.hylograph.net/code-explorer
polyglot.purescri.pt	Backend ecosystem showcase	GitHub Pages + link to live demos

Relationships: Hylograph is the *what* (visualization libraries). Minard is the *proof* (real-world application built with Hylograph). Polyglot is the *how* (PureScript compiles everywhere). Many Polyglot demos use Hylograph for visualization. Minard gains architectural enforcement features that are a prerequisite for the ShapedSteer intensive.

What follows: once all three ship, the ShapedSteer intensive begins (~8 weeks). Minard's architectural enforcement features are used to develop ShapedSteer with continuous structural validation. See [ShapedSteer/docs/plan-for-a-plan.md](#).

Part 1: hylograph.net (Library Suite)

1.1 Libraries to Publish

Rename from `purescript-psd3-*` to `purescript-hylograph-*`:

Library	Current	New Name	Demo	Status
selection	psd3-selection	hylograph-selection	Hylograph Explorer	Ready
simulation	psd3-simulation	hylograph-simulation	Force Playground	Ready
layout	psd3-layout	hylograph-layout	Layout Gallery	Ready
graph	psd3-graph	hylograph-graph	Honeycomb Puzzle	Ready
music	psd3-music	hylograph-music	Anscombe's String Quartet	Ready
transitions	psd3-transitions	hylograph-transitions	(part of selection demo)	Merge into selection?

Additional libraries (keep separate):

Library	Current	New Name	Notes
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Library	Current	New Name	Notes
transitions	psd3-transitions	hylograph-transitions	Animation/transition helpers
optics	psd3-optics	hylograph-optics	Lens-based tree traversal
d3-kernel	psd3-d3-kernel	hylograph-d3-kernel	D3 impl of simulation API
simulation-core	psd3-simulation-core	hylograph-simulation-core	Abstract simulation API
simulation-halogen	psd3-simulation-halogen	hylograph-simulation-halogen	Halogen integration
wasm-kernel	psd3-wasm-kernel	hylograph-wasm-kernel	Rust/WASM impl (separate build)
canvas	psd3-canvas	hylograph-canvas	Evaluate: keep if used

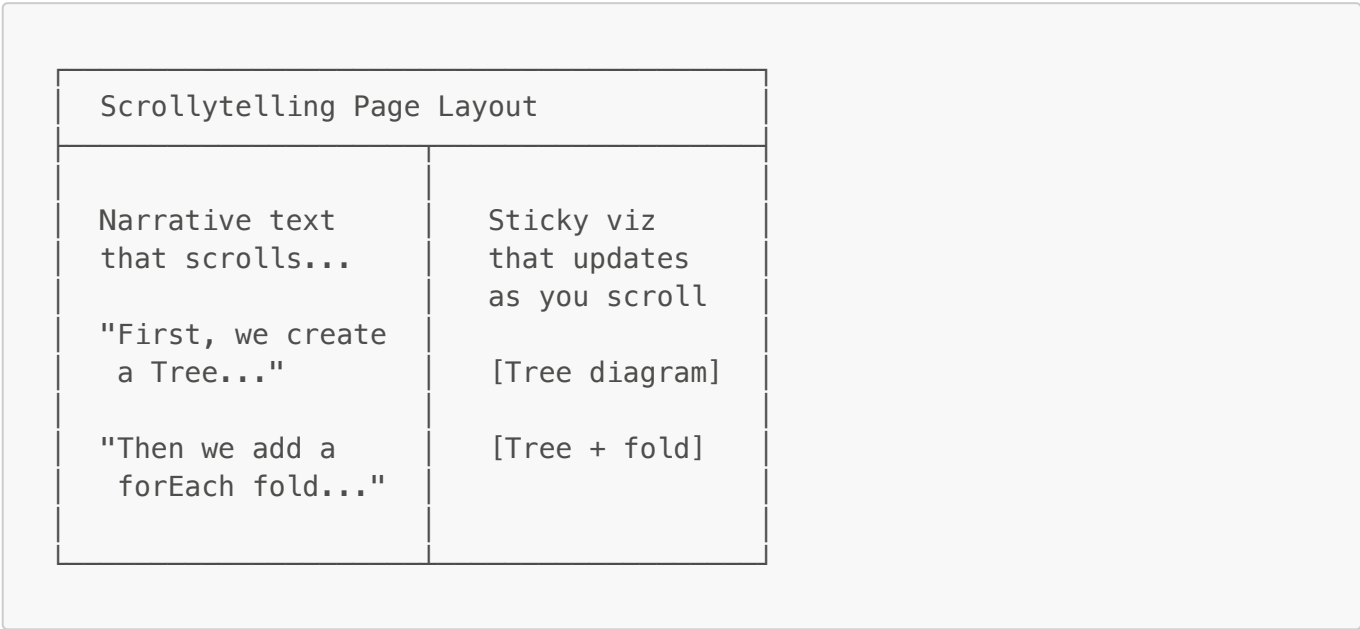
1.2 Domain Structure

hylograph.net/ Visualization for PureScript"	# Landing page: "Declarative
├── libs/	# Library documentation
│ ├── selection/	# Library page + Hylograph Explorer
│ ├── simulation/	# Library page + Force Playground
│ ├── layout/	# Library page + Layout Gallery
│ ├── graph/	# Library page + Honeycomb Puzzle
│ └── music/	# Library page + Anscombe's String
Quartet	
├── docs/	# Scrollytelling documentation
│ ├── getting-started/	# Quick start guide
│ ├── tutorials/	# Step-by-step tutorials
│ └── simpsons-paradox/	# Worked example: data-driven Halogen
component	
│ ├── how-to/	# Task-oriented guides
│ └── reference/	# API reference (Antora?)
├── dashboard/	# Interactive dashboard
│ └── (component playground, live examples)	
└── guide/	# Interactive HATS tour (hylograph-
guide)	
live.hylograph.net/ MacMini)	# Backend-required demos (Tunnel to
├── code-explorer/	# Corrode Expel
├── sankey/	# Arid Keystone (full version)
└── tidal/	# Tilted Radio (if ready)

Key insight: Simpson's Paradox is a *worked example* in the documentation, not a standalone demo. It shows "how to build a data-driven Halogen visualization component with Hylograph."

1.3 Documentation Approach: Scrollytelling

The hylograph.net documentation uses **scrollytelling** - narrative-driven pages where visualizations respond to scroll position:



Documentation sections:

- **Getting Started:** Install, hello world, first visualization
- **Tutorials:** Step-by-step builds (Simpson's Paradox is here)
- **How-To:** Task-oriented ("How to add interactivity", "How to animate")
- **Reference:** API docs, type signatures (possibly Antora-generated)

1.4 Dashboard

A live **dashboard** at hylograph.net/dashboard/ provides:

- Component playground (try HATS in browser)
- Example gallery with source code
- Performance benchmarks (D3 vs WASM)
- Build status of all libraries

This is where the current MacMini dashboard concept evolves to.

1.5 Static Demos (Cloudflare Pages)

Demo	Source	Notes
Hylograph Explorer	showcases/hylograph-app	Interactive HATS structure viewer
Force Playground	site/website (route)	Already in website
Layout Gallery	showcases/allergy-outlay	Move to layout lib site

Demo	Source	Notes
Honeycomb Puzzle	showcases/psd3-honeycomb	Move to graph lib site
Anscombe's String Quartet	psd3-anscombe-quartet	Move to music lib site
Optic Menagerie	showcases/emptier-coinage	Keep standalone or merge
Morphism Zoo	showcases/psd3-prim-zoo-mosh	Static, whimsical
WASM Force Demo	showcases/wasm-force-demo	Proves WASM backend works

1.6 Hylograph Release Tasks

Phase 1: Demo Consolidation

- [] Move Layout Gallery (allergy-outlay) → layout lib site
- [] Move Honeycomb (psd3-honeycomb) → graph lib site
- [] Move Anscombe → music lib site, rename "Anscombe's String Quartet"
- [] Move Simpson's Paradox → docs/tutorials/

Phase 2: Infrastructure

- [] Verify hylograph.net DNS active in Cloudflare
- [] Set up Cloudflare Pages project
- [] Create cloudflare-build Makefile target
- [] Configure Cloudflare Tunnel on MacMini
- [] Test end-to-end deployment

Phase 3: Extract to GitHub (flat repos under afcondon)

- [] Create purescript-hylograph-selection repo
- [] Create purescript-hylograph-simulation repo
- [] Create purescript-hylograph-layout repo
- [] Create purescript-hylograph-graph repo
- [] Create purescript-hylograph-music repo
- [] Create purescript-hylograph-transitions repo
- [] Create purescript-hylograph-optics repo
- [] Create purescript-hylograph-simulation-core repo
- [] Create purescript-hylograph-d3-kernel repo
- [] Create purescript-hylograph-wasm-kernel repo
- [] Create purescript-hylograph-simulation-halogen repo

Phase 4: Rename & Update

- [] Update all module names (PSD3.* → Hylograph.*)
- [] Update all spago.yaml package names
- [] Update all internal imports across repos
- [] Update documentation references

Phase 5: Publish

- [] Publish to PureScript Registry (in dependency order)
- [] Update website branding (PSD3 → Hylograph)
- [] Build scrollytelling documentation
- [] Launch dashboard

Phase 6: Documentation

- [] Getting Started guide
- [] Simpson's Paradox tutorial (scrollytelling)
- [] How-To guides
- [] API reference (Antora or custom)
- [] Test end-to-end
- [] Update website branding (PSD3 → Hylograph)

Phase 4: Documentation

- [] Update Antora docs with new names
- [] Add "Try it live" links to each library page
- [] Create getting started guide
- [] Record demo videos (optional)

Part 2: polyglot.purescri.pt (Ecosystem Showcase)

2.1 What It Demonstrates

The *Polyglot PureScript* project proves PureScript can target multiple backends:

Backend	Technology	Demo
JavaScript	Standard (browser + Node)	Everything
Python	PurePy compiler	Grid Explorer, Embedding Explorer
Erlang	Purerl compiler	Tilted Radio (Tidal)
Lua	PsLua compiler	Scuppered Ligature (edge router)
WASM	Rust interop	WASM Force Demo

2.2 Showcase Inventory

Tier 1: Flagship Demos (Backend-powered)

Name	Anagram	Backend	Frontend	Status	Notes
Code Explorer	Corrode Expel	Node.js	Halogen	80%	Real-world app potential
Grid Explorer	(Hypo- Punter)	PurePy	Halogen	90%	Great PurePy demo
Embedding Explorer	(Hypo- Punter)	PurePy	Halogen	90%	Great PurePy demo
Tidal Editor	Tilted Radio	Purerl	Halogen	70%	Algorave demo
Sankey Editor	Arid Keystone	(optional)	Halogen	85%	Build management potential

Tier 2: Static Showcases (No backend)

Name	Anagram	Technology	Status	Notes
Optic Menagerie	Emptier Coinage	JS	100%	Lens visualization
Morphism Zoo	Prim Zoo Mosh	JS	100%	Recursion schemes
Simpson's Paradox	-	JS	90%	Statistical visualization how-to
Layout Gallery	Allergy Outlay	JS	100%	Move to hylograph
WASM Force	-	Rust+JS	100%	WASM proof

Tier 3: Infrastructure (Not user-facing)

Name	Anagram	Technology	Notes
Edge Router	Scuppered Ligature	PsLua	Proves Lua backend works

2.3 External Projects to Reference

Project	Location	Notes
purescript-python-new	Separate repo	PurePy compiler itself
purescript-lua fork	GitHub	PRs pending (golden tests needed)
purescript-build	TBD	Build systems à la carte
purescript-diagrams	TBD	Haskell port
purescript-linear	TBD	Haskell port
purescript-machines	TBD	Haskell port

2.4 Domain Structure

polyglot.purescri.pt/	# "PureScript Runs Everywhere"
├ index.html	# Hero: backends diagram, quick links
├ backends/	# One page per backend
│ └ python/	# PurePy: compiler, ecosystem, demos
│ └ erlang/	# Purerl: OTP, websockets, Tidal
│ └ lua/	# PsLua: Nginx, edge computing
│ └ wasm/	# Rust interop: performance, WASM
├ showcases/	# Gallery cards linking to live demos
│ └ grid-explorer/	→ live.hylograph.net/grid-explorer
│ └ embedding-explorer/	→ live.hylograph.net/embedding-explorer
│ └ tidal-editor/	→ live.hylograph.net/tidal
│ └ edge-router/	# (no demo, just explanation)
└ ecosystem/	# Related projects
└ hylograph/	→ hylograph.net (visualization)

└─ diagrams/	→ purescript-diagrams repo
└─ linear/	→ purescript-linear repo
└─ machines/	→ purescript-machines repo

Key insight: polyglot.purescri.pt is a *showcase site* ("look what PureScript can do!"), not a documentation site. Documentation lives at hylograph.net.

Separation of concerns:

- **hylograph.net:** "How to build visualizations" (docs, tutorials, API reference)
- **polyglot.purescri.pt:** "PureScript compiles to X" (showcases, links, ecosystem map)

2.5 PR for purescript-domain

Add to `/Users/afc/work/afc-work/GitHub/purescript-domain/sites.yaml`:

```
polyglot:
  redirect: https://afcondon.github.io/polyglot-purescript
```

Or if using a custom domain later:

```
polyglot:
  mask: https://polyglot.example.com
```

2.6 Polyglot Release Tasks

Phase 1: Repository Organization

- └─ [] Decide: monorepo vs separate repos for showcases
- └─ [] Create polyglot-purescript landing page repo
- └─ [] Document relationship between polyglot and hylograph
- └─ [] Identify which showcases stay in PSD3-Repos vs move out

Phase 2: PurePy Ecosystem

- └─ [] Verify purescript-python-new is public and documented
- └─ [] Document ee-server and ge-server as examples
- └─ [] Create "Getting Started with PurePy" guide

Phase 3: Lua Ecosystem

- └─ [] Submit golden test PR to purescript-lua
- └─ [] Document Scupperd Ligature as Lua example
- └─ [] Create "PureScript in Nginx" guide

Phase 4: Deploy Landing Page

- └─ [] Create GitHub Pages site
- └─ [] Submit PR to purescript-domain/sites.yaml
- └─ [] Coordinate with Nick Saunders

```
Phase 5: Documentation
├─ [ ] Write overview of each backend
├─ [ ] Create architecture diagrams
├─ [ ] Add links to all live demos
```

Part 3: Minard / Code Explorer Release

3.1 What Ships

Minard is the flagship demo of the Hylograph ecosystem — a real-world application, not just a library showcase. It deserves its own release alongside the libraries.

Core product: Code cartography tool that visualizes PureScript codebases as interactive maps (treemaps, dependency graphs, declaration arc diagrams, type class grids).

Stack: Rust loader → DuckDB → PureScript/HTTPurple API → PureScript/Halogen/Hylograph frontend

What's ready today:

- Loader ingests source + compiler output + git history + spago.lock
- 13 interconnected visualization views
- 7 color modes (topological depth, git status, reachability, clusters, etc.)
- Type class grid, declaration detail, module signatures via Sigil
- Annotation system (AI + human review)
- Multi-project, multi-snapshot support

3.2 Architectural Enforcement Features (new for release)

These features turn Minard from a passive code map into an active architectural fitness function. They are also a prerequisite for the ShapedSteer intensive that follows.

Full details: [docs/kb/plans/minard-architectural-enforcement.md](#)

Feature	What it does	Effort
Layer config	<code>architecture.yml</code> at project root defines layers + module patterns	Half day
Schema + loader	New <code>architecture_layers</code> , <code>architecture_violations</code> tables; layer assignment during postload	1 day
Violation detection	Compute violations (upward imports) during postload	Half day
API endpoints	<code>/api/v2/architecture/{layers,violations,summary}</code>	Half day
Layer color mode	New <code>ArchitectureLayer</code> color mode in frontend	1 day

Feature	What it does	Effort
Violation overlay	Red edges, badges, status bar for violations	1-2 days
Module size warnings	LOC threshold badges on treemap	Half day
Drift tracking	Violations over time via snapshot comparison	1 day

Total: ~5-7 days. Steps 1-5 (~3 days) are minimum viable.

3.3 Minard Release Tasks

Phase 1: Core Polish

- [] Ensure all 13 views work end-to-end on a fresh codebase
- [] Fix any known bugs in loader/API/frontend
- [] Clean up UI (light theme, Swiss design consistency)
- [] Verify git integration works with real repos

Phase 2: Architectural Enforcement (5–7 days)

- [] architecture.yml config parsing in loader (Rust)
- [] Schema changes: architecture_layers, architecture_violations tables
- [] Layer assignment + violation computation in postload
- [] API endpoints: /api/v2/architecture/*
- [] ArchitectureLayer color mode in frontend
- [] Violation overlay (red edges, badges)
- [] Module size warning badges
- [] Drift tracking (snapshot comparison chart)
- [] Test: ingest ShapedSteer, verify 4 known violations detected

Phase 3: Documentation & Deployment

- [] README / getting started guide
- [] Pre-built loader binary for macOS (and Linux?)
- [] Docker profile in polyglot-deploy
- [] Deploy at live.hylograph.net/code-explorer
- [] Screenshot/demo GIF for landing page

3.4 Relationship to ShapedSteer

Minard with architectural enforcement is a **prerequisite** for the ShapedSteer intensive:

Ship Hylograph libs + Polyglot + Minard

↓

Write ShapedSteer architecture.yml

↓

Ingest ShapedSteer into Minard

↓

Begin ShapedSteer intensive with enforcement in place

↓

Reviewer agent checks Minard violations before approving merges

Part 4: Shared Infrastructure

3.1 MacMini Services

Current Docker stack serves both projects:

Service	Port	Used By
edge (Lua router)	80	Both
website	-	Hylograph
ce-backend	3000	Both
ce-frontend	8085	Both
ee-backend	5081	Polyglot
ee-frontend	8087	Polyglot
ge-backend	5082	Polyglot
ge-frontend	8088	Polyglot
tidal-backend	8083	Polyglot
tidal-frontend	8084	Polyglot
sankey	8089	Hylograph
hylograph	-	Hylograph
lib-*	-	Hylograph

3.2 Cloudflare Tunnel Configuration

```
# live.hylograph.net – Hylograph backend demos
ingress:
- hostname: live.hylograph.net
  path: /code-explorer/*
  service: http://localhost:8085
- hostname: live.hylograph.net
  path: /sankey/*
  service: http://localhost:8089
- hostname: live.hylograph.net
  path: /tidal/*
  service: http://localhost:8084

# live.polyglot.purescri.pt – Polyglot backend demos (or reuse
live.hylograph.net)
```

```

ingress:
- hostname: live.hylograph.net
  path: /grid-explorer/*
  service: http://localhost:8088
- hostname: live.hylograph.net
  path: /embedding-explorer/*
  service: http://localhost:8087

```

Part 5: Open Questions

Decisions Made (2026-01-29)

1. **Repository structure:** Flat set of repos on afcondon's GitHub

- All libraries become `purescript-hylograph-*` repos
- No monorepo - each library is independent
- Can transfer to org later if community interest grows

2. **Library consolidation:** Keep kernels separate

- `transitions` - keep separate (could merge later)
- `optics` - keep separate (could merge later)
- `d3-kernel` - keep separate (D3 wrapper for abstract simulation API)
- `simulation-core` - keep separate
- `wasm-kernel` - keep separate (Rust build complexity)
- Rationale: WASM kernel requires Rust toolchain, shouldn't force that on simulation users

3. **Demo embedding** (embed into library repos):

- Layout Gallery → layout lib
- Honeycomb → graph lib
- Force Playground → simulation lib
- Anscombe → music lib (rename "Anscombe's String Quartet")
- Hylograph Explorer → selection lib
- Optic Menagerie → optics lib

4. **Hylograph website extras:**

- Morphism Zoo → easter egg on hylograph.net (not a polyglot showcase)

5. **Separate showcase repos** (for polyglot.purescript.pt):

- Code Explorer (corrode-expel) - Node.js backend
- Grid Explorer (hypo-punter) - PurePy backend
- Embedding Explorer (hypo-punter) - PurePy backend
- Tidal Editor (tilted-radio) - Purerl backend
- Sankey Editor (arid-keystone) - future build tool, will merge with purescript-build
- Edge Router (scuppered-ligature) - separate repo, IS the Lua backend infrastructure

6. **Documentation:**

- Simpson's Paradox → docs/tutorials (worked example)

7. **Haskell library ports** (diagrams, linear, machines):

- Separate repos, not under hylograph umbrella
- Loosely part of polyglot.purescri.pt showcase

8. **Naming:**

- Keep anagram codenames as internal/fun names
- Use descriptive names publicly

Technical Notes

1. **Registry publishing order** (by dependency):

- graph → layout → selection → transitions → optics → simulation → music

2. **GitHub:** Start under afcondon, not a trapdoor (can transfer repos later)

Part 6: Timeline

Phase 1: Foundation (Week 1-2)

- ☐ Finalize library consolidation decisions
- ☐ Set up Cloudflare Pages for hylograph.net
- ☐ Set up Cloudflare Tunnel on MacMini
- ☐ Create polyglot-purescript landing page repo

Phase 2: Hylograph Release (Week 3-4)

- ☐ Rename all libraries psd3 → hylograph
- ☐ Update all imports and configurations
- ☐ Deploy to hylograph.net
- ☐ Test all demos work

Phase 3: Minard Release (Week 5-6)

- ☐ Core polish: all views working, known bugs fixed, light theme
- ☐ Architectural enforcement features (see Part 3, ~5-7 days)
- ☐ Test: ingest ShapedSteer, verify known violations detected
- ☐ Documentation: README, getting started, pre-built loader binary
- ☐ Deploy at live.hylograph.net/code-explorer

Phase 4: Polyglot Release (Week 7-8)

- ☐ Build and deploy polyglot.purescri.pt
- ☐ Submit PR to purescript-domain
- ☐ Coordinate with Nick Saunders
- ☐ Submit Lua golden test PR

Phase 5: Polish & Announce (Week 9-10)

- ☐ Final documentation review across all three
- ☐ Create announcement posts
- ☐ Post to PureScript Discord
- ☐ Post to Reddit/HN if desired

Phase 6: ShapedSteer Intensive Begins (Week 11+)

- ☐ Mac Mini dev server setup (see ShapedSteer plan-for-a-plan)
- ☐ Write architecture.yml, ingest into Minard
- ☐ Dry run: first feature with full workflow
- ☐ 8-week intensive begins

Appendix A: Current Directory Structure

```

PSD3-Repas/
├── visualisation libraries/
│   ├── purescript-psd3-selection      → hylograph-selection
│   ├── purescript-psd3-simulation    → hylograph-simulation
│   ├── purescript-psd3-layout        → hylograph-layout
│   ├── purescript-psd3-graph         → hylograph-graph
│   ├── purescript-psd3-music         → hylograph-music
│   ├── purescript-psd3-transitions   → merge or rename
│   ├── purescript-psd3-optics        → evaluate
│   ├── purescript-psd3-canvas        → evaluate
│   ├── purescript-psd3-d3-kernel     → evaluate
│   ├── purescript-psd3-simulation-core → merge into simulation
│   ├── purescript-psd3-simulation-halogen → keep
│   ├── purescript-psd3-wasm-kernel   → merge into simulation
│   └── psd3-anscombe-quartet        → move to music lib site
├── showcases/
│   ├── hylograph-app                 # Hylograph Explorer (NEW)
│   ├── hylograph-guide               # Interactive tutorial (WIP)
│   ├── corrode-expel                # Code Explorer
│   ├── hypo-punter                  # EE + GE (PurePy)
│   ├── psd3-tilted-radio             # Tidal (Purerl)
│   ├── psd3-arid-keystone            # Sankey Editor
│   ├── emptier-coinage              # Optic Menagerie
│   ├── psd3-prim-zoo-mosh            # Morphism Zoo
│   ├── allergy-outlay               # Layout Gallery → move to layout lib
│   ├── psd3-honeycomb               # Honeycomb → move to graph lib
│   ├── simpsons-paradox             # How-to example
│   ├── wasm-force-demo              # WASM proof
│   └── scuppered-ligature           # Edge router (Lua)
├── site/
│   ├── website/                    # Main hylograph.net
│   ├── lib-selection/              # Library landing pages
│   ├── lib-simulation/
│   └── lib-layout/

```

```
| | | lib-graph/
| | | lib-music/
| | | docs/           # Antora documentation
| | apps/
| | | shaped-steer    # Full Sankey/build app (separate)
```

Appendix B: Sites.yaml PR

```
# Add to purescript-domain/sites.yaml
polyglot:
  redirect: https://afcondon.github.io/polyglot-purescript
```

Once the page is ready, change to:

```
polyglot:
  mask: https://afcondon.github.io/polyglot-purescript
```